Example

WA, L1

$$f_X(x) = \begin{cases} \frac{e^x}{e^5 - 1} & \text{for } 0 \le x < 5 \\ 0 & \text{o.w.} \end{cases}$$

(1) Find CDF

$$F_X(x) = \int_{-\infty}^{x} \frac{e^t}{e^5 - 1}\, dt$$

$$= \frac{1}{e^5 - 1} \int_{-\infty}^{x} e^t\, dt \qquad \left(\text{factor out } \tfrac{1}{e^5-1} \text{ outside of integrand}\right)$$

$$= \frac{1}{e^5 - 1} \int_{0}^{x} e^t\, dt \qquad (\text{since no density exists at } (-\infty, 0))$$

$$= \frac{1}{e^5 - 1} [e^t]\Big|_0^x \qquad \left(\int e^t\, dt = e^t\right)$$

$$= \frac{1}{e^5 - 1} [e^x - e^0] \qquad (\text{Fundamental Theorem of Calculus})$$

$$= \frac{e^x - 1}{e^5 - 1} \quad \text{for } 0 \le x < 5$$

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{e^x - 1}{e^5 - 1} & \text{for } 0 \le x < 5 \\ 1 & \text{for } x \ge 5 \end{cases}$$

(2) $P(X<1) = F_X(1)$

$= \frac{e^1 - 1}{e^5 - 1}$

$\approx \boxed{0.0117}$

(3) $P(2<X<4) = F_X(4) - F_X(2)$

$= \frac{e^4 - 1}{e^5 - 1} - \frac{e^2 - 1}{e^5 - 1}$

$\approx 0.3636 - 0.0433$

$\approx \boxed{0.3203}$

(4) $P(X>2) = 1 - F_X(2)$

$= 1 - 0.0433$

$= \boxed{0.9567}$

(5) CDF → PDF

$$\frac{d}{dx} F_X(x) = \begin{cases} \frac{d}{dx}(0) \\ \frac{d}{dx}\left[\frac{e^x - 1}{e^5 - 1}\right] \\ \frac{d}{dx}(1) \end{cases} = \begin{cases} \frac{e^x}{e^5 - 1} & \text{for } 0 \le x < 5 \\ 0 & \text{otherwise} \end{cases}$$

$$\frac{d}{dx}\left(\frac{e^x - 1}{e^5 - 1}\right) = \frac{1}{e^5 - 1}\frac{d}{dx}(e^x - 1)$$

$$= \frac{1}{e^5 - 1}(e^x - 0)$$

$$= \frac{e^x}{e^5 - 1}$$